



COMSATS University Islamabad, Lahore Campus
Department of Computer Engineering

ASSIGNMENT NO 2- FALL 2025

Course Title:	Operating System			Course Code:	CSC323	Credit Hours:	3(2,1)
Course	Engr. Ahmad Mudassir			Program	BCE		
Semester:	5 th	Batch:	FA23-BCE	Section:	A,B	Date:	8 th December 2025
Deadline	11 th December 2025			Maximum Marks:	20		
Student's Name:				Reg. No.	CUI/	/LHR	

Important Instructions / Guidelines:

Note: Assignments must reflect each student's individual work. Any instance of plagiarism, duplication, or copying from other students or online sources will result in a zero (0) grade for all parties involved.

(Complex 2-Level Paging with Incomplete Page Tables)

A 64-bit system uses:

- Page size: 8 KB
- Page directory: 4096 entries
- Page table: 4096 entries per table
- Virtual address: 48 bits used

You have an incomplete page table dump:

PD[0x03A] -> PT @ frame 0x1F20

PD[0x07C] -> PT @ frame 0x00A0 (marked "lazy allocated")

PT[0x100] -> frame 0x0041 (present)

PT[0x205] -> frame 0x0ACD (dirty)

PT[0x3AB] -> INVALID

Given virtual address:

0x03A_205_1F4

Tasks:

- a) Break the virtual address into PD index, PT index, offset (show bit counts).
- b) Translate the virtual address step-by-step.
- c) Explain what happens if PT entry is "lazy allocated."
- d) Show what occurs if PT[0x205] has the dirty bit set but goes through eviction.
- e) Identify whether TLB must be flushed after this translation.(Justify)